D Y PATIL INTERNATIONAL UNIVERSITY
AKURDI PUNE

School of Computer Science, Engineering and Applications(SCSEA)

Academic Year: 2025-26 *B.Tech S.Y (Sem IV)*

Subject: Data Mining and Warehousing(CSE2103)

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Title: Data Exploration and Ethical Analysis

Aim:

To study and Perform basic exploratory data analysis on a dataset and identify ethical issues related to privacy, bias, and responsible data usage.

Experiment/Procedure:

1. Start Jupyter Notebook and import the required libraries: pandas & numpy.

2. Load the given dataset into a DataFrame .

3. Display First Few Records.

4. Check Dataset Information.

5. Describe the Dataset.

6. Generate Statistical Summary.

7. Ethical Analysis Section (Privacy Risk Detection)

8. Demographic Bias Detection.

9. Ethical Concerns

10. Responsible data usage check.

11. End of Analysis

Result:

In [1]:
```
#Tanishka Misal
#PRN : 20240802496

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

In [4]:
```
#Tanishka Misal
#PRN : 20240802496

#step1: Load the dataset
file_path = (r"C:\Users\lenovo\Downloads\Wearable_Health_Lifestyle_Ethics.csv")
df =  pd.read_csv(file_path)
print("Dataset loaded successfully!\n")
```

```
Dataset loaded successfully!
```

In [7]:
```
#Tanishka Misal
#PRN : 20240802496

#step2: basic data exploration
print("First 5 Records:\n")
print(df.head())

print("Dataset Shape(Rows , Columns)")
print(df.shape)

print("Column Information")
print(df.info())

print("Statistical Summary")
print(df.describe())
```

```
First 5 Records:

  User_ID  Age  Gender   BMI  Daily_Steps  Avg_Sleep_Hours  \
0   U1001   56    Male  31.4        16493              6.6
1   U1002   46  Female  21.2        18908              8.2
2   U1003   32    Male  30.9         9224              6.6
3   U1004   60  Female  31.6         5025              9.1
4   U1005   25  Female  19.6        19950              5.8

   Resting_Heart_Rate  Stress_Level  Water_Intake_Liters Smoking_Habit  \
0                  55             8                  3.8           Yes
1                  52             3                  1.9           Yes
2                  84             2                  0.7            No
3                  67             7                  1.1           Yes
4                  60             8                  4.6            No

       Region Income_Level Health_Risk_Category  Insurance_Premium  \
0  Semi-Urban       Medium                 High           16338.10
1  Semi-Urban       Medium                 High           18167.37
2       Rural          Low                 High           22531.97
3  Semi-Urban       Medium                 High           15300.18
4       Urban         High                 High           18792.82
```

```
  Data_Shared_With_ThirdParty
0                         Yes
1                          No
2                         Yes
3                         Yes
4                         Yes
Dataset Shape(Rows , Columns)
(1000, 15)
Column Information
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 15 columns):
 #   Column                        Non-Null Count  Dtype
---  ------                        --------------  -----
 0   User_ID                       1000 non-null   object
 1   Age                           1000 non-null   int64
 2   Gender                        1000 non-null   object
 3   BMI                           1000 non-null   float64
 4   Daily_Steps                   1000 non-null   int64
 5   Avg_Sleep_Hours               1000 non-null   float64
 6   Resting_Heart_Rate            1000 non-null   int64
 7   Stress_Level                  1000 non-null   int64
 8   Water_Intake_Liters           1000 non-null   float64
 9   Smoking_Habit                 1000 non-null   object
 10  Region                        1000 non-null   object
 11  Income_Level                  1000 non-null   object
 12  Health_Risk_Category          1000 non-null   object
 13  Insurance_Premium             1000 non-null   float64
 14  Data_Shared_With_ThirdParty   1000 non-null   object
dtypes: float64(4), int64(4), object(7)
memory usage: 117.3+ KB
None
Statistical Summary
                Age           BMI   Daily_Steps  Avg_Sleep_Hours  \
count  1000.000000  1000.000000   1000.000000      1000.000000
mean     40.986000    27.927900  10387.620000         6.385800
std      13.497852     6.902965   5544.191237         2.023609
min      18.000000    16.000000   1025.000000         3.000000
25%      29.000000    22.200000   5333.250000         4.600000
50%      42.000000    28.000000  10928.000000         6.300000
75%      52.000000    33.825000  15113.250000         8.100000
max      64.000000    40.000000  19991.000000        10.000000


       Resting_Heart_Rate  Stress_Level  Water_Intake_Liters  \
count         1000.000000   1000.000000          1000.000000
mean            73.956000      4.918000             2.753200
std             14.158903      2.567463             1.283486
min             50.000000      1.000000             0.500000
25%             62.000000      3.000000             1.700000
50%             73.000000      5.000000             2.800000
75%             86.000000      7.000000             3.800000
max             99.000000      9.000000             5.000000

       Insurance_Premium
count        1000.000000
mean        17667.787720
std          7994.572709
min          4000.630000
25%         10230.465000
50%         18642.210000
75%         24390.690000
max         29992.580000
```

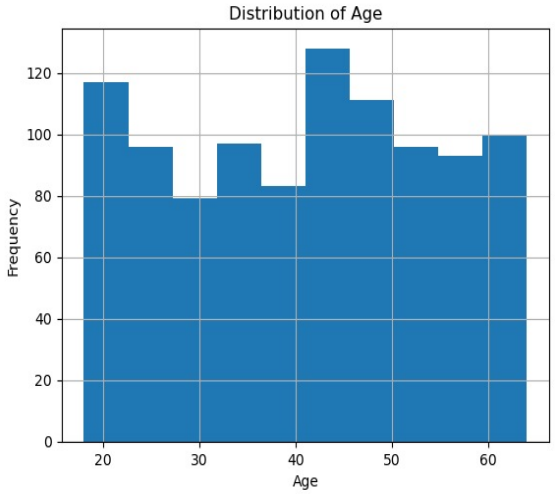
In [8]:

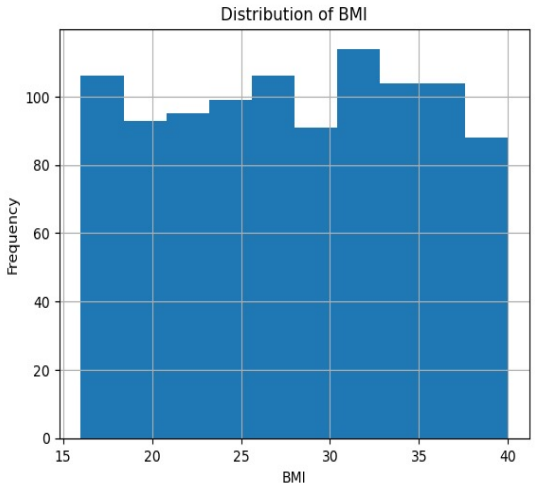
```
#Tanishka Misal
#PRN : 20240802496

#step3 :Missing values analysis
print("Misssing Values Per Column:")
print(df.isnull().sum())
```

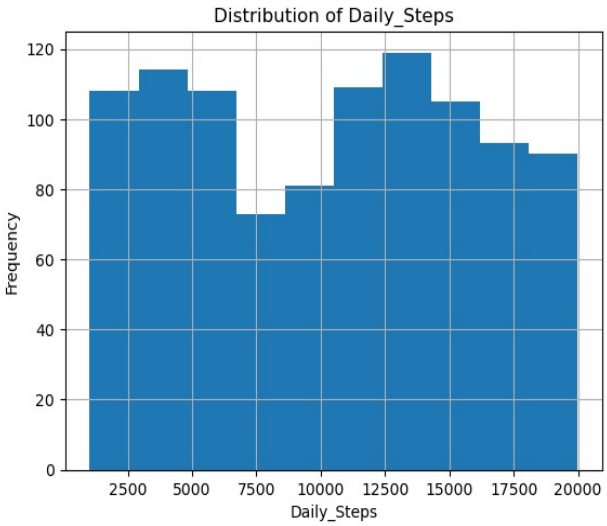
```
Misssing Values Per Column:
User_ID                        0
Age                            0
Gender                         0
BMI                            0
Daily_Steps                    0
Avg_Sleep_Hours                0
Resting_Heart_Rate             0
Stress_Level                   0
Water_Intake_Liters            0
Smoking_Habit                  0
Region                         0
Income_Level                   0
Health_Risk_Category           0
Insurance_Premium              0
Data_Shared_With_ThirdParty    0
dtype: int64
```

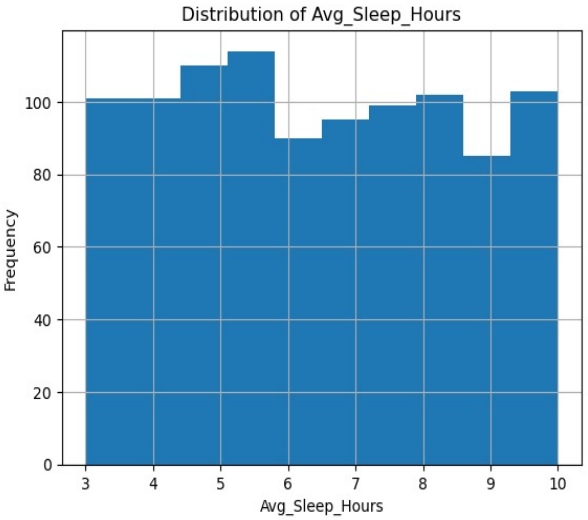
In [11]:

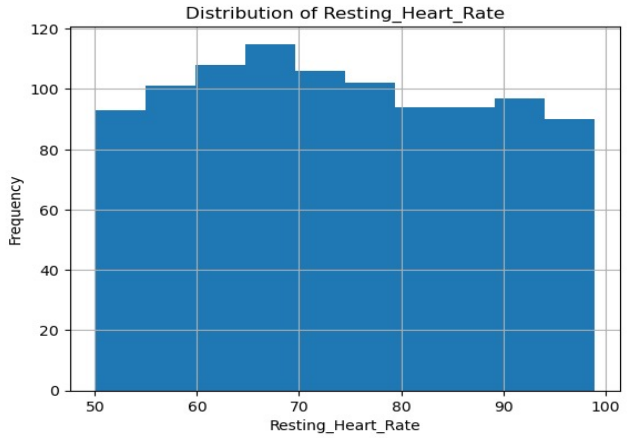
```
#Tanishka Misal
#PRN : 20240802496

#step4:Visual Exploration
#plot numeric distribution
numeric_cols = df.select_dtypes(include=np.number).columns
for col in numeric_cols:

    df[col].hist()
    plt.title(f"Distribution of(col)")
    plt.xlabel(col)
    plt.ylabel("Frequency")
    plt.show()
```

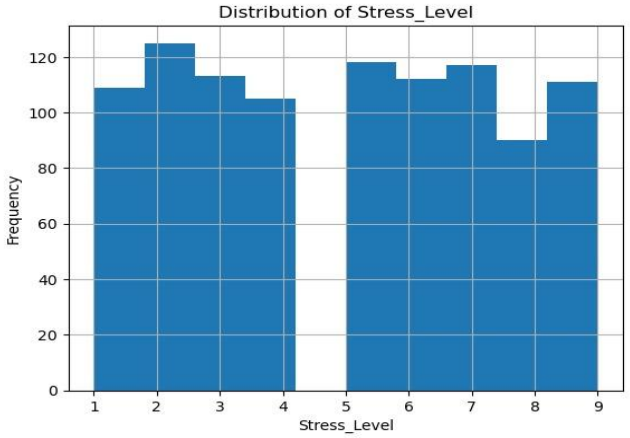


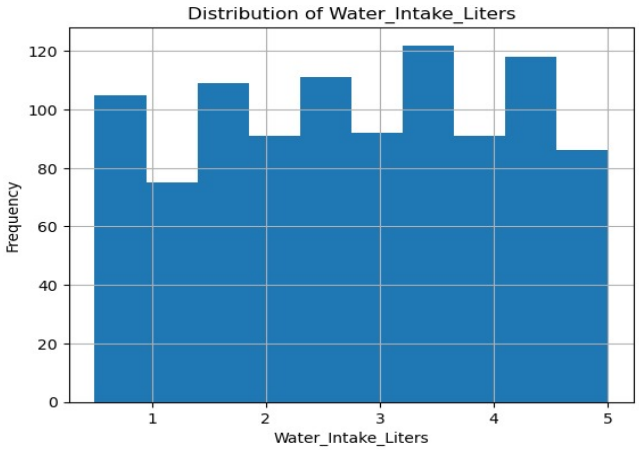


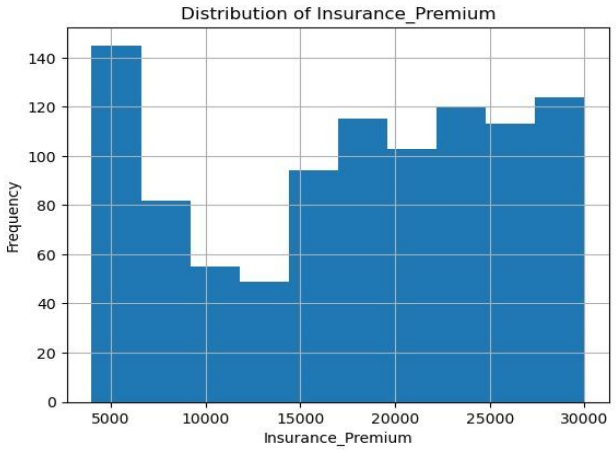








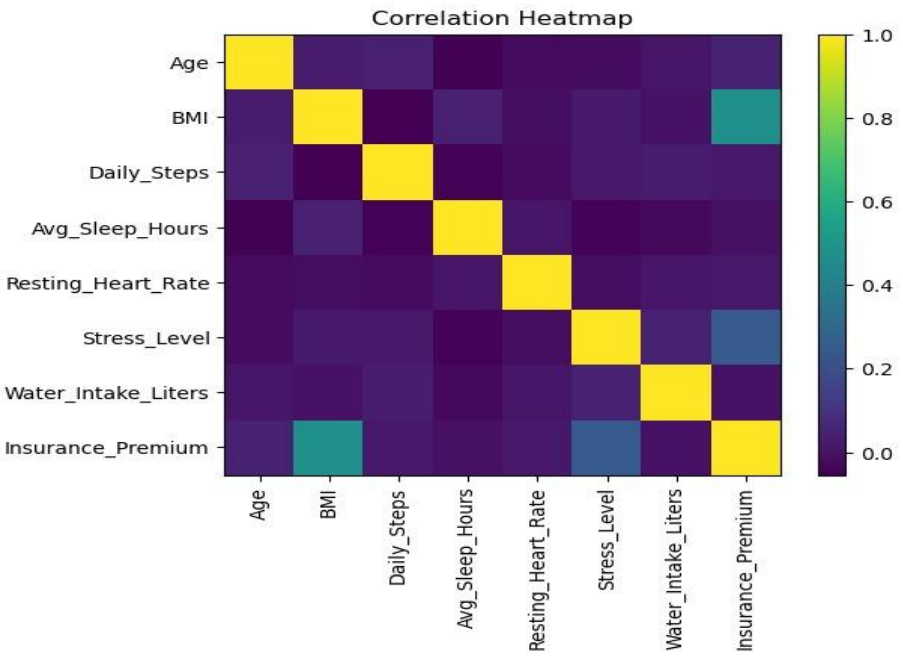




In [15]:

```python
#Tanishka Misal
#PRN : 20240802496

#step5:correlation analysis
print("Correlation Matrix")
correlation_matrix = df.corr(numeric_only=True)
print(correlation_matrix)

plt.figure()
plt.imshow(correlation_matrix)
plt.colorbar()
plt.title("Correlation Heatmap")
plt.xticks(range(len(correlation_matrix.columns)), correlation_matrix.columns,rotation = 90)
plt.yticks(range(len(correlation_matrix.columns)), correlation_matrix.columns)
plt.show()
```

```
Correlation Matrix
                        Age       BMI  Daily_Steps  Avg_Sleep_Hours  \
Age                1.000000  0.024641     0.042104        -0.047440
BMI                0.024641  1.000000    -0.053853         0.043808
Daily_Steps        0.042104 -0.053853     1.000000        -0.044813
Avg_Sleep_Hours   -0.047440  0.043808    -0.044813         1.000000
Resting_Heart_Rate -0.023903 -0.015268    -0.023734         0.010763
Stress_Level      -0.017191  0.022749     0.016745        -0.045019
Water_Intake_Liters  0.009698 -0.001735     0.030804        -0.029944
Insurance_Premium  0.049038  0.475127     0.016927        -0.008144

                    Resting_Heart_Rate  Stress_Level  Water_Intake_Liters  \
Age                           -0.023903     -0.017191             0.009698
BMI                           -0.015268      0.022749            -0.001735
Daily_Steps                   -0.023734      0.016745             0.030804
Avg_Sleep_Hours                0.010763     -0.045019            -0.029944
Resting_Heart_Rate             1.000000     -0.013344             0.010760
Stress_Level                  -0.013344      1.000000             0.047832
Water_Intake_Liters            0.010760      0.047832             1.000000
Insurance_Premium              0.016438      0.247485            -0.003123

                    Insurance_Premium
Age                          0.049038
BMI                          0.475127
Daily_Steps                  0.016927
Avg_Sleep_Hours             -0.008144
Resting_Heart_Rate           0.016438
Stress_Level                 0.247485
Water_Intake_Liters         -0.003123
Insurance_Premium            1.000000
```




```python
#Tanishka Misal
#PRN : 20240802496

#step6:ethical analysis section
#privacy risk detection

sensitive_keywords = ["name","email","phone","address","location","gps","id","dob"]
privacy_risk = []

#check for sensitive columns
for col in df.columns:
    for keyword in sensitive_keywords:
        if keyword.lower() in col.lower():
            privacy_risk.append(col)

if privacy_risk:
    print("PRIVACY RISK DETECTED!!")
    print("Sensitive columns found: ", privacy_risk)
else:
    print("No obvious direct personal identifiers detected.")

#basic detection(demographic skew check)
demographic_cols = ["Gender","Age","Region"]

for col in demographic_cols:
    if col in df.columns:
        print(f"\nDistribution of {col}:")
        print(df[col].value_counts(normalize=True))


print("Ethical Concerns:")
print("If one demographic group dominates the dataset,")
print("AI models trained on it may show biased predictions.")

#Responsible data usage check
print("Responsible Data Usage Considerations:")
print("- Was user consent taken?")
print("- Is the data anonymized?")
print("- Is the data used strictly for health improvement ?")
print("- Is it shared with the third parties?")

print("\n========= END OF ANALYSIS ========")

```

```
PRIVACY RISK DETECTED!!
Sensitive columns found:  ['User_ID']

Distribution of Gender:
Gender
Female        0.491
Male          0.467
Non-binary    0.042
Name: proportion, dtype: float64

Distribution of Age:
Age
43    0.033
45    0.030
50    0.030
52    0.028
54    0.027
64    0.027
49    0.025
22    0.025
18    0.025
62    0.025
56    0.025
41    0.025
42    0.024
20    0.024
40    0.024
19    0.024
61    0.023
25    0.023
23    0.023
53    0.023
29    0.022
34    0.022
39    0.022
47    0.021
33    0.021
36    0.020
28    0.020
46    0.019
38    0.019
21    0.019
31    0.019
51    0.018
27    0.018
58    0.018
37    0.018
26    0.018
30    0.018
```


```
47    0.021
33    0.021
36    0.020
28    0.020
46    0.019
38    0.019
21    0.019
31    0.019
51    0.018
27    0.018
58    0.018
37    0.018
26    0.018
30    0.018
```


```
32    0.018
59    0.017
57    0.017
35    0.016
48    0.016
44    0.016
55    0.016
24    0.014
63    0.013
60    0.012
Name: proportion, dtype: float64

Distribution of Region:
Region
Urban         0.348
Rural         0.346
Semi-Urban    0.306
Name: proportion, dtype: float64
Ethical Concerns:
If one demographic group dominates the dataset,
AI models trained on it may show biased predictions.
Responsible Data Usage Considerations:
- Was user consent taken?
- Is the data anonymized?
- Is the data used strictly for health improvement ?
- Is it shared with the third parties?

========= END OF ANALYSIS ========
```

Conclusion:

In this practical, data exploration and ethical analysis were performed on the wearable health lifestyle dataset using Python. The dataset was first loaded successfully using the pandas library, and initial exploration techniques such as viewing the first few rows, checking dataset information, and generating statistical summaries were carried out. These steps helped in understanding the structure, content, and characteristics of the dataset.